Profiling Instruction Routing in Vision-Language-Action Models: A Cross-Architecture Diagnostic Framework

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Abstract

Vision-Language-Action (VLA) models fine-tune large language models on robot demonstration data, yet how training reshapes the internal flow of instruction information remains poorly understood. We present a systematic diagnostic framework for token-level representational profiling of decoder-only VLAs, validated across three architectures—OpenVLA (Llama-2-7B), TracVLA (Phi-3-mini), and NORA (Qwen2.5-VL-3B)—and four LIBERO task suites (10 matrix cells). Our framework combines factorial experimental designs (training condition × instruction presence), six complementary probing methods, and counterfactual perturbation analysis. Applying this framework, we identify two consistent cross-architecture phenomena: (1) R^2 *exit collapse*—a substantial drop in action-predictive information at exit layers—emerges in all six VLM-to-VLA transition cells ($\Delta = 0.156$–0.455) but in none of the four VLA-to-VLA fine-tuning cells ($\Delta = 0.033$–0.086), revealing it as a one-time reorganization signature; (2) trained VLAs exhibit counterfactual sensitivity to instruction perturbation (Cohen's $d \geq 0.45$), though metric incommensurability between trained and untrained conditions limits cross-condition quantification. An NLP fine-tuning control ($\Delta R^2 = 0.000$ across three chat models) confirms these effects are specific to action-oriented training. We also show that L1 attention head counts previously attributed to VLA training are in fact base-model-dependent (e.g., Phi-3-Vision: 32/32 heads in both trained and untrained conditions). While our framework provides the first systematic profiling toolkit for VLA representational analysis, the primary findings—exit collapse as a VLM-to-VLA transition artifact and counterfactual sensitivity without behavioral grounding—represent descriptive characterization rather than mechanistic or actionable insights for VLA design.

1 Introduction

Vision-Language-Action (VLA) models have emerged as a promising paradigm for robotic manipulation, fine-tuning pretrained vision-language models to generate continuous robot actions from visual observations and natural language instructions [**??**]. Despite rapid adoption, the internal mechanisms by which these models process and route instruction information during action generation remain opaque. Prior behavioral studies have identified visual dominance [**?**] and instruction underutilization [**?**] in VLA outputs, but have not decomposed the representational roles of different token positions or explained how fine-tuning reshapes information flow.

Understanding instruction routing is important for VLA development: if instruction information is systematically discarded or poorly integrated, architectural modifications could improve task conditioning. Conversely, if routing patterns are trivial consequences of architectural constraints (e.g., causal masking), then research effort should focus elsewhere.

We present a diagnostic framework for token-level representational profiling of decoder-only VLAs. Our framework employs: (i) factorial experimental designs crossing training condition (VLA-trained, VLM-pretrained, random initialization) with instruction presence; (ii) six complementary analysis methods (RSA, CKA, R^2 action probes, MLP probes, delta probes, classification); (iii) counterfactual perturbation analysis with multiple instruction modifications; and (iv) attention head profiling across all layers.

We validate this framework across three VLA architectures with distinct LLM backbones—OpenVLA (Llama-2-7B), TracVLA (Phi-3-mini), and NORA (Qwen2.5-VL-3B)—spanning four LIBERO task suites. Our main findings are:

1. **R^2 exit collapse** is a consistent VLM-to-VLA transition marker: 6/6 VLM→VLA cells show substantial collapse ($\Delta = 0.156$–0.455), while 0/4 VLA→VLA cells exceed the $\Delta > 0.10$ threshold.
2. **Counterfactual sensitivity** to instruction perturbation is present in all trained VLAs ($d \geq 0.45$), but ablation of instruction-dominant attention heads does not degrade action quality (ΔMSE ≈ 0), indicating a dissociation between representational sensitivity and functional impact.
3. **L1 attention head counts** are base-model-dependent, not training-induced: Phi-3-Vision exhibits 32/32 instruction-dominant heads even without VLA training.
4. **Metric dependence**: CKA reverses RSA ordering for trained vs. untrained representations, indicating that geometric and kernel-based similarity measures capture different aspects of representational structure.

We position these contributions as providing the first systematic profiling toolkit for VLA representational analysis, while acknowledging that the empirical findings are primarily descriptive: exit collapse appears trivially expected when a language model is adapted to produce actions, and counterfactual sensitivity lacks behavioral validation through closed-loop evaluation.

2 Related Work

Vision-Language-Action Models. VLAs adapt pretrained VLMs to generate robot actions by fine-tuning on demonstration data. Representative architectures include RT-2 [**?**], which discretizes actions into language tokens; OpenVLA [**?**], which uses a Llama-2-7B backbone with a DinoSigLIP vision encoder; TracVLA, built on Phi-3-mini; and NORA, which employs Qwen2.5-VL-3B with grouped-query attention. These models vary in vision encoder, LLM backbone, and attention mechanism, providing a natural testbed for cross-architecture analysis.

Representational Analysis of Transformers. Probing classifiers [**??**], representational similarity analysis (RSA) [**?**], and centered kernel alignment (CKA) [**?**] are standard tools for analyzing transformer representations. Recent work has highlighted that probe accuracy alone may not indicate functional use of information [**?**], motivating multi-method approaches. The dissociation between probing results and functional impact is particularly relevant to our counterfactual sensitivity findings.

Instruction Following in Embodied Agents. Grant et al. [**?**] demonstrated visual dominance in VLA behavior, finding that instructions contribute minimally to action selection. Haon et al. [**?**] characterized instruction underutilization in deployment. Our work complements these behavioral studies by examining the representational mechanisms underlying instruction processing, though we note that our analysis is open-loop and does not include closed-loop behavioral validation.

3 Method: Diagnostic Framework for VLA Profiling

3.1 Overview

Our framework profiles how instruction information flows through a decoder-only VLA by comparing representations under systematically varied conditions. The core design is a factorial matrix crossing *training condition* (trained VLA, pretrained VLM/VLA, random initialization) with *instruction presence* (original instruction, empty, shuffled, wrong-task, paraphrased).

For a VLA with L transformer layers, we extract hidden-state representations $\mathbf{h}_l^{(t)} \in \mathbb{R}^d$ at each layer $l \in \{0, \ldots, L\text{–}1\}$ for each token position t. We distinguish two aggregation strategies: *image_mean* (averaging over all image token positions) and *last_preaction* (the final token position before action decoding).

3.2 Analysis Methods

We employ six complementary methods, each capturing a different aspect of representational structure:

1. **Representational Similarity Analysis (RSA)**: Spearman correlation between pairwise representation distances and action distances across conditions.

2. **Centered Kernel Alignment (CKA)**: Linear CKA between layer representations and action structure, providing a kernel-based complement to RSA.
3. **R^2 action probes**: Ridge regression from layer representations to 7-DoF action vectors, evaluated via Monte Carlo cross-validation (MCCV, $n = 1000$).
4. **MLP action probes**: Two-layer MLP probes for nonlinear action prediction.
5. **Delta probes**: Probing for training-induced representational changes.
6. **Cross-instruction classification**: 10-way classification of which instruction was active, given same-frame representations.

3.3 Counterfactual Perturbation Protocol

To assess instruction sensitivity, we run the VLA on identical visual inputs with five instruction conditions: (i) original instruction, (ii) empty string, (iii) shuffled tokens, (iv) wrong-task instruction, and (v) paraphrased instruction. For trained models that can decode actions, we measure output deviation via Cohen's d in action space. For untrained models (VLM baselines) that cannot produce valid actions, we measure cosine distance in hidden-state space. This methodological asymmetry limits cross-condition comparisons (Section 4.4).

3.4 Architectures and Datasets

We evaluate three VLA architectures:

- **OpenVLA**: Llama-2-7B backbone (32 layers, 32 MHA heads), DinoSigLIP vision encoder, trained on Open X-Embodiment (970K episodes) then fine-tuned on LIBERO.
- **TracVLA**: Phi-3-mini backbone (32 layers, 32 MHA heads), Phi-3-Vision encoder.
- **NORA**: Qwen2.5-VL-3B backbone (36 layers, 16 heads with 2 KV groups via GQA).

Each is evaluated on up to four LIBERO suites: Goal, Spatial, Object, and Long-Horizon. This yields a 3×4 matrix of which 10/12 cells are filled (NORA lacks Object and Long checkpoints).

4 Experiments

4.1 Token-Level Specialization (C1 Framework)

Cross-Instruction Classification. We evaluate whether instruction identity can be decoded from different token positions across layers, using 10-way classification on same-frame, different-instruction conditions (10 tasks × 50 timesteps × 10 demos per condition).

Table 1: Cross-instruction classification accuracy across conditions and layers. Image-mean tokens remain at chance (10%) due to causal masking. Trained and base models saturate quickly; random-init decays in deep layers.

Condition	L1 Acc	L31 Acc	LOOCV	Interpretation
Trained	99.6%	100%	100.0±0.0	Ceiling at all layers
Llama-2 base	80.2%	100%	—	LM pretraining provides separability
Untrained (random)	100%	82%	84.8±11.6	Signal decays in deep layers
Image mean	10%	10%	—	Chance: causal mask guarantee

Table 1 reveals that 10-way instruction classification on 4096-dimensional features is trivially easy: even random-initialization achieves 100% at L1. The framework's value lies not in this ceiling result but in the three-condition comparison: trained models maintain 100% through all 32 layers, Llama-2 base saturates by L5, and random initialization decays from 100% to 82%, indicating that structured pretrained weights (whether VLA-trained or not) preserve instruction signals through depth.

Table 2: 2×2 factorial design: RSA of image tokens under training $\times$ instruction conditions. Both factors produce null effects ($\Delta < 0.005$), as predicted by causal masking and frozen encoder constraints. Bootstrap CI includes zero for both effects.

Condition	RSA	Reviewer Verdict
Trained + Instruction	0.488	Expected: frozen encoder dominates
Trained − Instruction	0.489	Expected: causal mask blocks
Untrained + Instruction	0.485	Expected: same encoder
Untrained − Instruction	0.480	Expected: both factors null
Training effect		$\Delta = 0.003$ (n.s.)
Instruction effect		$\Delta = 0.004$ (n.s.)

Instruction Ablation and 2×2 Factorial Design. We cross training condition with instruction presence to test whether image-token representations are affected by instruction content.

The 2×2 null result (Table 2) confirms that image-token RSA is invariant to both training and instruction content, with all four conditions converging to the 0.48–0.49 band. This is structurally predicted: the frozen SigLIP encoder produces identical image features regardless of downstream LLM state, and causal masking prevents instruction tokens from influencing earlier image tokens.

Random Instruction Control. We extend the instruction invariance test with four instruction conditions: normal ($\rho = 0.489$), random tokens ($\rho = 0.484$), shuffled tokens ($\rho = 0.483$), and empty ($\rho = 0.472$). The maximum absolute deviation is $\Delta = 0.017$, confirming that image-token RSA is invariant to instruction content, consistent with the causal mask guarantee. Combined with the 2×2 result and the direct instruction ablation ($\Delta = 0.004$), these three analyses verify the same structurally guaranteed null. We include them as a sanity check rather than an empirical finding.

4.2 Vision Encoder Geometry

Table 3: Encoder quality measured by RSA and R^2 action probes. DINO (self-supervised, no language) achieves highest RSA and competitive R^2, while VL-pretrained encoders (SigLIP, CLIP) do not outperform. Language supervision appears unnecessary for action-relevant geometric structure.

Encoder	RSA	R^2	Language Supervised?	Implication
DINO cls	0.692	0.634	No	Best: language unnecessary
SigLIP+Projector	0.615	0.630	Yes	VL pretraining inferior to DINO
CLIP ViT-L	0.557	0.587	Yes	Worst VL encoder
VLA trained (L12 peak)	0.628	—	—	LLM processing approaches DINO
DINO patch mean	0.631	—	No	Consistent with cls token

Cross-Encoder RSA and Pretraining Controls. Table 3 presents cross-encoder comparisons including a CLIP-ViT-L pretraining control. The finding that DINO outperforms both VL-pretrained encoders in RSA ($\rho = 0.692$ vs. 0.615 for SigLIP, 0.557 for CLIP) challenges the assumption that language-supervised pretraining benefits action-relevant representation quality.

SigLIP+Projector RSA Scope. Table 4 shows that the frozen SigLIP+Projector provides the highest RSA ($\rho = 0.615$), and LLM processing degrades geometric alignment by $\Delta = 0.127$ regardless of whether the LLM is trained or randomly initialized. This decomposition is limited to OpenVLA's DinoSigLIP encoder; TracVLA (Phi-3-Vision) and NORA (InternViT) use different vision encoders not tested here.

Table 4: SigLIP+Projector RSA decomposition (OpenVLA only). The frozen vision encoder provides the highest geometric alignment ($\rho = 0.615$); LLM processing degrades it to $\rho = 0.488$, a drop of $\Delta = 0.127$.

Source	RSA	Architecture	Note
SigLIP+Projector	0.615	OpenVLA only	Frozen encoder
VLA trained (peak)	0.488	OpenVLA only	LLM degrades -0.127
VLA untrained (peak)	0.485	OpenVLA only	Random-init LLM similar
DINO	0.388	Standalone	Patch-mean reference

Branch-Split Analysis. Within OpenVLA's fused DinoSigLIP backbone, the DINOv2 branch ($\rho = 0.619$) and SigLIP branch ($\rho = 0.621$) contribute nearly identically to the fused output ($\rho = 0.634$), which slightly exceeds the post-projector value ($\rho = 0.615$). This architecture-specific decomposition demonstrates the framework's sub-encoder analysis capability but does not generalize to other VLA encoders.

Table 5: Metric ordering reversal between RSA and CKA. The two metrics give opposite rankings for trained vs. untrained representations, indicating that conclusions about action-relevant structure are metric-dependent.

Metric	Ordering	Implication
RSA (geometric)	SigLIP > Trained > Untrained > DINO	Training degrades geometry
CKA (global)	Untrained > Trained > SigLIP > DINO	Training preserves structure
R^2 (functional)	$\text{Trained}_{\text{lastpre}}$ > DINO $\approx$ Proj > $\text{Trained}_{\text{img}}$ > U	Training enables routing

Per-Layer RSA and CKA Divergence. The per-layer RSA curve for OpenVLA shows an inverted-U pattern: RSA rises from projector output ($\rho = 0.615$) to a peak at L9 ($\rho_{\text{trained}} = 0.488$, $\rho_{\text{untrained}} = 0.485$) before declining. The rise phase is small ($\Delta\rho = 0.016$) with a confidence interval that includes zero. Critically, CKA reverses the RSA ordering entirely (Table 5): under CKA, untrained > trained, whereas under RSA, trained > untrained. This dissociation means that claims about representational structure are metric-dependent—a result we report transparently as it affects interpretation throughout the paper.

Output Head Analysis. The vision-projector cosine similarity is exactly 1.0 between trained and untrained conditions, confirming the encoder is frozen as implemented. LLM weight relative changes are uniformly distributed (mean = 1.316, range 1.22–1.37), reflecting pretrained-vs-random weight differences rather than fine-tuning effects. This verification confirms implementation correctness but does not constitute a finding.

Table 6: Trained vs. untrained differences across six analysis methods. Deltas are uniformly small, consistent with the frozen-encoder prediction. CKA reverses the RSA direction, breaking full-method consistency.

Method	Trained$-$Untrained Δ	Note
RSA (geometric)	$+0.003$	Expected: frozen encoder
R^2 action probe	$+0.015$	Within noise
CKA (global)	-0.006	**Reversal**: undermines RSA
MLP action probe	$+0.045$	Small routing effect
Delta probe	$+0.011$	Same encoder $\rightarrow$ similar
Classification	$+0.001$	Trivially easy for both

Six-Method Consistency Check. Six methods (Table 6) confirm that trained and untrained VLAs have nearly identical intermediate representations ($\Delta < 0.05$ for all methods). This is structurally expected

given the shared frozen encoder. The CKA reversal ($\Delta = -0.006$, opposite sign to RSA's +0.003) prevents claiming full multi-method consistency.

4.3 R^2 Exit Collapse

Table 7: R^2 action probe profiles across conditions (OpenVLA, LIBERO-Goal). Exit collapse ($\Delta = 0.256$) appears only in the VLA-trained condition. NLP fine-tuning (Llama-2-chat) shows flat noise floor, confirming action-specificity.

Condition	Peak R^2 (Layer)	Exit R^2 (L32)	Δ	Interpretation
VLA trained	0.678 (L13)	0.422	0.256	Action head consumes info
OXE untrained	0.609 (L32)	0.609	0.000	No collapse (VLA→VLA)
Llama-2-chat (D100)	0.156 (all)	0.156	0.000	Flat noise floor
Random init	0.640 (L0)	∼0 (L10+)	pathological	Not valid control

Full-Layer R^2 Scan (OpenVLA). Table 7 shows the full-layer R^2 profile for OpenVLA on LIBERO-Goal. The trained model exhibits a clear peak at L13 ($R^2 = 0.678$) followed by decline to $R^2 = 0.422$ at exit (L32), yielding $\Delta = 0.256$. The OXE-pretrained untrained condition shows no collapse ($\Delta = 0.000$, peak at L32 itself), consistent with the VLA-to-VLA interpretation developed in Section 4.3. Random initialization produces pathological profiles unsuitable as a control.

Table 8: NLP fine-tuning controls across three LLM backbones. All show flat $R^2 \approx 0.155$ (noise floor) with $\Delta = 0.000$, confirming that exit collapse requires action-oriented training, not general fine-tuning.

Model	Backbone	ΔR^2	Profile	Note
Llama-2-7b-chat	Llama-2-7B	0.000	Flat 0.155	No action info anywhere
Phi-3-mini-instruct	Phi-3-mini	0.000	Flat 0.155	Same noise floor
Qwen2.5-3B-instruct	Qwen2.5-3B	0.000	Flat 0.155	Same noise floor
VLM→VLA (TracVLA)	Phi-3-mini	0.156–0.294	Exit collapse	VLA-specific
VLA→VLA (OpenVLA)	Llama-2-7B	0.033–0.086	Mild decline	No new collapse

NLP Fine-Tuning Control. Three NLP-fine-tuned models sharing the same backbones as our VLAs all show flat $R^2 \approx 0.155$ across all layers (Table 8), with $\Delta R^2 = 0.000$. The $R^2 \approx 0.155$ reflects the noise floor from between-task variance (10 tasks → Ridge regression captures task identity → ∼0.155 R^2). This control confirms exit collapse is specific to action-oriented training, ruling out generic fine-tuning as a confound. However, it does not address whether exit collapse is a trivially expected consequence of any domain adaptation that introduces an action decoding head.

Table 9: Cross-benchmark R^2 comparison for OpenVLA. LIBERO-Spatial replicates the pattern from Goal: small trained-vs-untrained deltas consistent with VLA→VLA transition producing minimal representational change.

Benchmark	Δ (imgmean)	Δ (lastpre)	Note
LIBERO-Goal	0.015	0.069	Only architecture tested
LIBERO-Spatial	0.023	0.040	Confirms VLA→VLA minimal

Cross-Benchmark Replication (LIBERO-Spatial). LIBERO-Spatial (Table 9) replicates the LIBERO-Goal pattern within OpenVLA: trained-vs-untrained deltas are small (0.023 for image_mean, 0.040 for

last_preaction), consistent with VLA→VLA transitions producing minimal additional representational reorganization. This is within-architecture replication across task suites, not cross-architecture validation.

Table 10: R^2 exit collapse matrix: VLM→VLA vs. VLA→VLA transitions across architectures and task suites. VLM→VLA transition produces substantial collapse in 6/6 cells ($\Delta = 0.138$–0.691); VLA→VLA fine-tuning produces no cell above the $\Delta > 0.10$ threshold (0/4 cells, $\Delta = 0.033$–0.086). MCCV v3 corrects prior train-test leakage.

Transition	Architecture / Suite	Δ (peak−exit)	Collapse?
VLM→VLA	TracVLA / Goal	0.228	Yes
VLM→VLA	TracVLA / Spatial	0.138	Yes
VLM→VLA	TracVLA / Object	0.294	Yes
VLM→VLA	TracVLA / Long	0.198	Yes
VLM→VLA	NORA / Goal	0.691	Yes
VLM→VLA	NORA / Spatial	0.302	Yes
VLA→VLA	OpenVLA / Goal	0.086	No
VLA→VLA	OpenVLA / Spatial	0.037	No
VLA→VLA	OpenVLA / Object	0.037	No
VLA→VLA	OpenVLA / Long	0.033	No

Cross-Architecture R^2 Exit Collapse. The cross-architecture exit collapse matrix (Table 10) reveals a clean binary separation. All six VLM-to-VLA cells (TracVLA and NORA, where untrained baselines are pure VLMs) show substantial R^2 exit collapse ($\Delta = 0.138$–0.691). All four VLA-to-VLA cells (OpenVLA, where the untrained baseline is already OXE-pretrained) show no substantial collapse ($\Delta = 0.033$–0.086, all below the $\Delta > 0.10$ threshold).

This binary separation indicates that exit collapse is a one-time reorganization occurring at the VLM-to-VLA transition boundary, not a cumulative effect of continued training. The magnitude varies substantially within VLM→VLA cells (4.4× range, 0.138–0.691), with NORA/Goal ($\Delta = 0.691$) as a clear outlier at 3× the median, suggesting architecture-dependent magnitude despite direction-consistent occurrence.

Table 11: Practical utility of exit collapse as a diagnostic tool. While directionally consistent, its one-time nature and magnitude variability limit quantitative diagnostic applications.

Question	Answer	Practical Value
Does model show VLM→VLA transition?	Yes, 6/6 cells	Low: training history known
Can it track ongoing optimization?	No, 0/4 VLA→VLA	Zero: one-time marker
Is magnitude calibrated?	No, 4.4× range	Low: cannot compare across arch.
Does NLP control exclude confound?	Yes, D100 $\Delta R^2 = 0$	Defensive: doesn't add value

Diagnostic Value Assessment.

4.4 Counterfactual Sensitivity Analysis

Cross-Architecture Counterfactual Results. Table 12 presents the cross-architecture counterfactual sensitivity results. All trained VLAs show substantial sensitivity to instruction perturbation ($d \geq 0.45$ or $\cos \geq 0.5$). However, 5 of 7 cross-architecture comparisons use incommensurable metrics: trained models produce actions (measured by Cohen's d in action space), while untrained VLMs cannot generate valid actions (measured by cosine distance in hidden space). Only OpenVLA (where both trained and untrained are action-producing VLAs) allows fair comparison, showing trained sensitivity 12–25% higher than untrained. NORA's trained condition saturates ($\cos = 1.0$), suggesting a ceiling artifact.

Table 12: Counterfactual sensitivity across architectures. Only 2/7 architecture-suite pairs allow fair trained-vs-untrained comparison (same metric). 5/7 pairs use incommensurable metrics (Cohen's d in action space vs. cosine distance in hidden space).

Arch	Suite	Trained	Untrained	Comparable?	Fair Δ
OpenVLA	Goal	$d = 1.182$	$d = 1.048$	Yes	+0.134 (12%)
OpenVLA	Spatial	$d = 1.173$	$d = 0.936$	Yes	+0.237 (25%)
TracVLA	Goal	$d = 1.524$	$\cos = 0.153$	No	Incommensurable
TracVLA	Spatial	$d = 1.333$	$\cos = 0.180$	No	Incommensurable
TracVLA	Object	$d = 1.212$	$\cos = 0.185$	No	Incommensurable
TracVLA	Long	$d = 1.473$	$\cos = 0.173$	No	Incommensurable
NORA	Goal	$\cos = 1.0$	$\cos = 0.081$	Partial	Saturated

Table 13: Gap between what counterfactual analysis measures and what would establish functional significance. No closed-loop behavioral data is available.

CF Shows	Reviewer Expects	Gap
Output direction changes ($d > 0.5$)	Task success rate drops	No closed-loop data
Empty > wrong > shuffled > para	Severity $\rightarrow$ failure rate	No behavioral grounding
ΔMSE ≈ 0 vs ground-truth	Action quality impact	MSE ≈ 0: no quality impact

Missing Behavioral Validation. A critical limitation is the absence of behavioral validation (Table 13). Counterfactual perturbations change output direction ($d = 0.65$–1.5) but do not degrade ground-truth action quality ($\Delta\text{MSE} = -0.001$, $p = 0.52$). This dissociation suggests that instruction information modulates output direction without affecting action quality—consistent with an information-computation dissociation [**?**], but also consistent with instruction routing being functionally redundant.

Table 14: Spearman $\rho = 1.0$ between token novelty and Cohen's d, indicating that CF sensitivity tracks lexical/positional distance rather than semantic content. Trajectory-level d inflates token-level d by $\sim 4\times$ via temporal autocorrelation.

Perturbation	Token Novelty	Token d	Trajectory d	Note
Empty	1.0	0.968	3.438	Max novelty $\rightarrow$ max effect
Wrong	0.83	0.844	3.798	High novelty
Paraphrased	0.47	0.697	2.131	Low novelty $\rightarrow$ low effect
Shuffled	0.0	0.625	2.595	Same tokens, different order

Token Overlap and Surface-Level Sensitivity. Token overlap analysis (Table 14) reveals a perfect rank correlation ($\rho = 1.0$) between token novelty and Cohen's d. This indicates that CF sensitivity is primarily driven by lexical/positional distance rather than semantic distance. The shuffled condition (same tokens, different order, $d = 0.625$) shows that positional information contributes beyond pure lexical overlap, but the overall pattern suggests surface-level rather than semantic instruction processing. Trajectory-level d values (2.1–3.8) inflate token-level values by $\sim 4\times$ through temporal autocorrelation.

Head Ablation and Functional Specificity. Head ablation (Table 15) reveals that instruction-dominant heads lack functional specificity. Ablating 24 instruction-dominant heads produces Cohen's $d = 0.651$, only $1.08\times$ that of random 24-head ablation ($d = 0.601$). The per-head effect is nearly identical across groups ($d \approx 0.025$). Most critically, ground-truth MSE is unchanged ($\Delta\text{MSE} = -0.001$, $p = 0.52$), meaning that removing 75% of "instruction routing" heads does not degrade action prediction quality. This result

Table 15: Head ablation results. Instruction-dominant heads show no disproportionate functional role: their per-head effect ($d = 0.027$) is comparable to random heads ($d = 0.025$), and ground-truth action quality is unaffected (ΔMSE $= -0.001$, $p = 0.52$). Specificity ratio of 1.08× is negligible.

Ablation Group	24h d	Per-head d	GT ΔMSE	Note
Instr-dominant 24	0.651	0.027	-0.001 ($p = 0.52$)	Direction change, not quality
Random 24 (avg)	0.601	0.025	similar	Any 24 heads → similar
Image-dominant 24	0.580	0.024	similar	No group specificity
Control 8 (random)	0.410	0.051	similar	Per-head effect *larger*

challenges the interpretation that attention-based instruction routing is functionally important for action generation.

4.5 Cross-Architecture Attention Analysis

Table 16: L1 instruction-dominant head counts across architectures. Training does not increase head counts; in TracVLA, it reduces them. Head counts reflect base-model language pretraining, not VLA fine-tuning.

Architecture	Base Model	Trained	Untrained	Training Δ
OpenVLA	Llama-2-7B	24/32	0/32 (random)	Artifact: random has no LM
OpenVLA	OXE-pretrained	24/32	25/32 (OXE)	−1: OXE saturated
TracVLA	Phi-3-mini	27–30/32	32/32 (VLM)	−2 to −5: training reduces
NORA	Qwen2.5-VL-3B	12–13/16	13–14/16	−1: no training effect

L1 Attention Heads: Base-Model Artifact (D090). Table 16 presents our most significant self-corrective finding. The L1 instruction-dominant head count—previously the most visually striking result (0→9→24 heads)—is shown to be base-model-dependent rather than training-induced. TracVLA's untrained VLM baseline (Phi-3-Vision) already has 32/32 instruction-dominant heads at L1; VLA training actually *reduces* this to 27–30. NORA's untrained baseline has 13–14/16 heads, unchanged by training. The original 0→24 gradient in OpenVLA reflects random-init-vs-pretrained differences, not VLA training effects.

Table 17: Status of originally claimed contributions after the D090 base-model control. The experimental framework survives; most empirical findings are removed or weakened.

Originally Claimed	Post-D090 Status	Remaining Value
L1 routing gradient (0→9→24)	Removed: base-model artifact	Zero
Bookend attention pattern	Weakened: entry from base	Exit engagement only
R^2 exit collapse	Retained: training-induced	One-time transition marker
CF sensitivity	Retained: training-induced	Unfair baselines
Experimental framework	Intact	Primary contribution

Contribution Audit Post-D090.

4.6 Three-Tier Routing Gradient

We observe a three-tier structure (Table 18): VLM bases (Tier-1) show no exit collapse and low CF sensitivity; the OXE-pretrained VLA (Tier-2) shows strong CF sensitivity but no exit collapse; task-fine-tuned VLAs (Tier-3) show exit collapse only when originating from VLMs. However, Tier-2 contains only a single model (openvla-7b), preventing statistical validation of a gradient. L1 head counts are non-monotonic across

Table 18: Three-tier routing gradient. Tier-2 has only $N = 1$ data point (openvla-7b), preventing statistical establishment of a gradient. L1 head counts are non-monotonic across tiers (Tier-1 Phi-3 = 32 > Tier-2 = 25 ≈ Tier-3 = 24–30).

Tier	N Models	Key Metrics	Limitation
1 (VLM)	3+	L1: 0–32; CF < 0.2; no collapse	Heterogeneous (0 vs 32)
2 (General VLA)	**1 only**	L1: 25; CF $d = 1.05$; no collapse	$N = 1$: cannot establish gradient
3 (Task VLA)	∼10 cells	L1: 13–30; CF $d > 0.45$; collapse if VLM-origin	Split by origin

tiers (Phi-3-Vision at Tier-1 has 32/32 heads, exceeding all Tier-3 models), further undermining the gradient narrative. We report this as a preliminary observation requiring additional intermediate-tier checkpoints.

4.7 Synthesis: What VLA Training Contributes

Table 19: Predicted reviewer assessment under worst-case interpretation. Mean score of 4.67/10 reflects strong methodology but thin post-D090 findings and missing behavioral validation.

Reviewer	Score	Primary Concern	Verdict
R1 (behavioral)	4	No closed-loop validation	Reject
R2 (methodology)	6	Framework solid; findings thin	Weak Accept
R3 (generalizability)	4	Magnitude variance; baselines	Reject
Mean	4.67	2 reject + 1 weak accept	Likely Reject

After the D090 base-model control, VLA training's unique contributions reduce to two effects, both with significant caveats:

1. **R^2 exit collapse**: Consistently present in VLM→VLA transitions (6/6 cells) but absent in VLA→VLA (0/4). Clean binary separation, confirmed not generic fine-tuning (D100). However, this may be a trivially expected consequence of adapting a language model to produce actions—the exit layer necessarily reorganizes to support action decoding.
2. **Counterfactual sensitivity**: Present in all trained conditions ($d \geq 0.45$). However, incommensurable baselines prevent quantifying the training amplification in 5/7 pairs; token-overlap $\rho = 1.0$ suggests surface-level sensitivity; and ΔMSE ≈ 0 indicates no functional consequence.

The experimental framework itself—factorial design, six complementary methods, cross-architecture validation protocol—represents the most robust surviving contribution, though methodology contributions at robotics venues traditionally receive lower scores than systems or behavioral results.

5 Discussion

Honest Assessment of Findings. Our framework was designed to discover instruction routing mechanisms in VLAs. What it primarily discovered is that many apparently training-induced patterns are actually inherited from base models (L1 attention heads), that the most consistent training effect (exit collapse) is a one-time transition artifact, and that instruction sensitivity ($d \geq 0.45$) tracks lexical distance ($\rho = 1.0$ with token novelty) without affecting action quality (ΔMSE ≈ 0). These are scientifically honest findings, but they do not provide actionable insights for VLA design.

The Diagnostic Utility Question. We positioned exit collapse as a diagnostic tool, but its practical utility is limited: it tells practitioners "this model was once a VLM that became a VLA"—information they already possess from the training pipeline. It cannot track ongoing optimization (0/4 VLA→VLA cells), and its magnitude varies 4.4× across architectures, precluding quantitative cross-architecture comparison.

Metric Dependence. The CKA-RSA dissociation (Table 5) is a fundamental challenge: if geometric (RSA) and kernel-based (CKA) measures disagree on ordering, which reflects "true" action-relevant structure? Our framework includes both metrics as a principled response, but this also means single-metric conclusions are unreliable. Future work should develop metric-independent criteria for representational quality.

Missing Behavioral Grounding. The absence of closed-loop evaluation is the most significant limitation for a robotics venue submission. Our counterfactual analysis shows that instruction perturbations change output direction ($d = 0.65$–1.5) without degrading ground-truth error ($\Delta\text{MSE} \approx 0$). Whether these directional changes affect task success rates in deployment remains unknown. Behavioral validation—running perturbed-instruction policies in simulation and measuring success rates—is critical future work.

Framework Value. Despite thin empirical findings, the framework itself has reusable value. The factorial design (training × instruction), multi-method analysis suite, and cross-architecture validation protocol can be applied to new VLA architectures as they are developed. The D090 finding—that the framework's own analysis revealed a previously claimed training effect was actually a base-model artifact—paradoxically demonstrates the framework's analytical power, though it simultaneously weakens the paper's empirical contributions.

6 Conclusion

We presented a systematic diagnostic framework for profiling instruction routing in Vision-Language-Action models, validated across three architectures (OpenVLA, TracVLA, NORA) and four LIBERO task suites. Our framework identified two consistent cross-architecture training effects: R^2 exit collapse as a one-time VLM-to-VLA transition marker (6/6 VLM→VLA cells, $\Delta = 0.138$–0.691; 0/4 VLA→VLA cells), and counterfactual sensitivity to instruction perturbation ($d \geq 0.45$). An NLP fine-tuning control ($\Delta R^2 = 0.000$) confirms action-specificity. However, exit collapse may be a trivially expected consequence of domain adaptation; counterfactual sensitivity tracks surface-level lexical features ($\rho = 1.0$) without functional consequence ($\Delta\text{MSE} \approx 0$); and L1 attention patterns reflect base-model pretraining rather than VLA training (D090). The framework's primary value lies in its reusable methodology for VLA representational analysis, enabling future work to distinguish training-induced from inherited representational properties and to ground representational findings in behavioral outcomes.

Limitations. (1) No closed-loop behavioral validation of counterfactual sensitivity. (2) Incommensurable metrics between trained and untrained conditions in 5/7 CF comparison pairs. (3) Three-tier routing gradient based on $N = 1$ at the intermediate tier. (4) RSA and CKA disagree on representational ordering. (5) Exit collapse magnitude varies 4.4× across architectures.